MIL-F-17690D
26 October 1984
SUPERSEDING
MIL-F-17690C
18 April 1980

MILITARY SPECIFICATION

FILTER ASSEMBLY, FLUID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers filter assemblies, fluid used for the removal of insoluble impurities from power plant turbine oil and diesel engine lubricating oil (additive-type).

1.2 Classification. Filter assemblies shall be of the following types and classes, as specified (see 6.2.1):

Type I - Single filter, assembly.
Type II - Dual filter, assemblies.

Class 1 - Without heater.
Class 2 - With heater.

1.3 Definitive military specification part number. The definitive military specification part number for items described in this specification will be identified as shown in 6.5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4330

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SPECIFICATIONS

FEDERAL

F-F-351 - Filter and Filter Elements, Fluid, Pressure, Lubricating Oil, Bypass and Full Flow.
GG-G-76 - Gage, Pressure and Vacuum, Dial Indicating, (For Air, Steam, Oil, Water, Ammonia, and Choro-Fluoro Hydrocarbon Gases, and Compressed Gases).
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.
MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514 - Plate Identification, Instruction and Marking, Blank.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts: Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures For Packaging of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking And Waterproofing, With Appropriate Test Methods.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B31.1- Power Piping.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII.
Pressure Vessels, Division I.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice For Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The filter assembly shall include the following: Shell, cover, removable filter elements, gages, piping, valves, and additional equipment required for the satisfactory operation of the unit. The equipment shall be complete with all accessories necessary, and shall be ready for operation when connected to the oil lines, drain, and, for class 2, electric power, steam or hot water supplies.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish a filter assembly for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The filter assembly shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the filters being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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3.4 Interchangeability. All filter assemblies of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Design. The filter assemblies shall be designed to withstand strains, jars, vibrations, and other conditions incidental to shipping, storage, installation, and operation and to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment. The filter assemblies shall be furnished complete in all details including those not specifically mentioned herein. The filter assemblies shall be designed for full-flow or by-pass operation, as specified (see 6.2.1).

3.6.1 Capacity. The design rate of flow of oil in gallons per hour (gph) at 200 Seconds Saybolt Universal (SSU) viscosity through the filter element shall be not less than that specified (see 6.2.1).

3.6.2 Components. Type I filter assembly shall consist of one shell and associated parts. Type II filter assemblies shall consist of two shells and associated parts arranged for parallel operation.

3.6.2.1 Shell. The shell of the filter element container shall be of pressed steel or welded steel construction, and shall enclose, support, and protect all elements. Compartments shall be provided for dirty oil and filtered oil. Supports shall be provided for base or side mountings adequate to support the unit in service. Tie-bolts, springs, or equivalent means shall hold the filter elements in recesses or other devices for receiving and loading the elements. Unless otherwise specified (see 6.2.1), shell shall be designed and constructed for 100 pound-force per square inch gage (psig) working pressure and shall conform to the ASME Code for Pressure Vessels, Section VII, Pressure Vessels, Division I.

3.6.2.2 Oil compartments. The compartment for dirty oil shall contain a sump for the collection of rejected particles. The inlet oil opening shall be above the sump. A drain connection shall lead from the sump. If the outlet oil connection is so placed that a space is left below the entrance level of the outlet which could serve as an oil sump, the drain connection shall be in the bottom of this space. The shell shall be so designed and constructed that dirt dropping from filter elements is prevented from entering the clean oil

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